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AI-Driven Salt Pan Mapping Using Multispectral Remote Sensing and GIS

District-Scale Mapping, Multispectral Index Extraction & Machine-Learning Benchmark

PROJECT METADATA & TEAM:

- Study Area: Thoothukudi (Tuticorin) District, Tamil Nadu, India
- Project Guide: Dr. Jenila Vincent | School of Computer Science and Engineering
- Student: Raafid Afraaz G. | Team Member: Farhanaaz R S
- Sensors: Copernicus Sentinel-2 MSI Level-2A & ESA WorldCover 2021
- Interactive Application: Thoothukudi Salt Pan Intelligence Dashboard



EXECUTIVE SUMMARY

Project Overview: AI-Driven Salt Pan Intelligence

Multi-stage geospatial workflow from raw Sentinel-2 L2A tiles to interactive dashboard

1. GEOSPATIAL PROBLEM

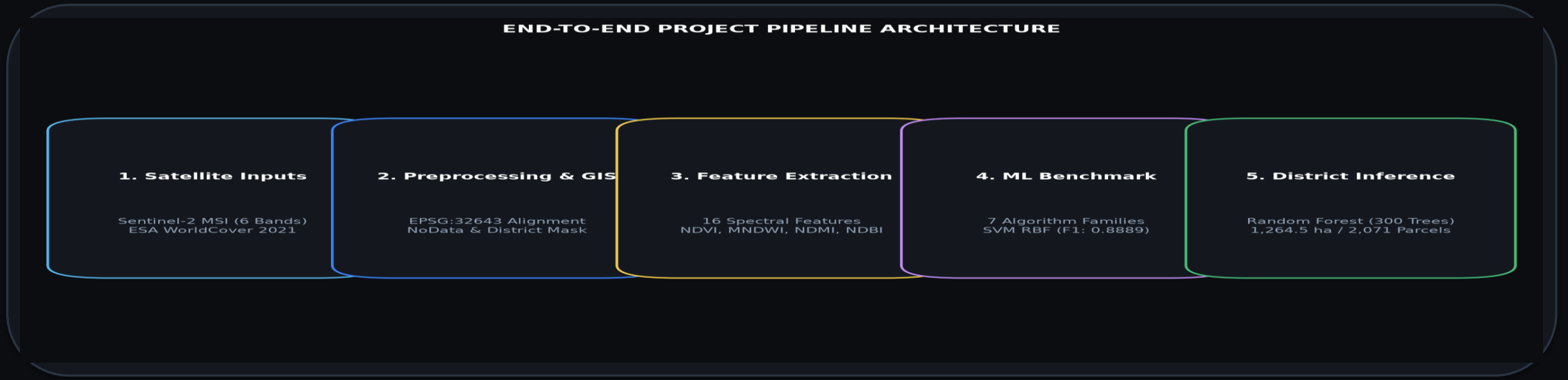
- Thoothukudi produces ~30% of India's marine salt
- Salt pans spectrally shift between dry brine crusts and open water reservoirs
- Severe confusion with coastal tidal flats, beaches & saline agricultural soils

2. ML & GIS APPROACH

- 16 engineered multispectral features (B02-B12, NDVI, MNDWI, NDMI, NDBI)
- 8,426 polygon training samples with 15.8:1 imbalance
- Rigorous 7-algorithm benchmark; SVM RBF won on F1 (0.8889)

3. VERIFIED OUTCOMES

- 1,264.52 ha final vector inventory across 2,071 parcels
- Strictly clipped to official district boundary (627 out-of-district purged)
- Full-stack React & MapLibre GL JS intelligence dashboard



PROBLEM FORMULATION

Problem Statement: Challenges in Salt Pan Delineation

Why traditional visual interpretation and simple thresholding fail in coastal salterns

THE MULTISPECTRAL CHALLENGE

- Spectral Inconstancy: During pre-crystallization, condenser pans contain deep brine that resembles shallow coastal water or wetlands.
- Crystallization Saturation: When crystallized, dry salt beds exhibit extreme visible reflectance (albedo > 0.8), mirroring white sand beaches and bare saline soils.
- Spatial Fragmentation: Salt pans are bounded by narrow earthen bunds (1-3 m wide), which are below native 10m/20m pixel resolution, causing mixed pixel artifacts.
- Cloud & Sun-Glint Confusion: High-reflectance cloud edges and coastal sun-glint introduce severe false positives in naive thresholding pipelines.
- Need for ML: Non-linear spectral band ratios (SWIR, NIR, MNDWI) and supervised classifiers are mandatory to achieve robust separation.

Challenge 1: High Intra-Class Spectral Variance

- Salt pans cycle between dark open water (Class 80 proxy) and blinding white crusts (Class 60/62 proxy). Single-band thresholds fail.

Challenge 2: Severe Class Imbalance (15.82 : 1)

- Salt pans represent less than 1% of Thoothukudi district's terrestrial surface. Standard classifiers collapse into majority non-salt prediction without balanced class weights.

Challenge 3: Manual Digitization Inefficiency

- District-scale manual digitization across 4,700 km^2 is slow, subjective, and unable to capture rapid seasonal expansion or abandonment.

CONTEXT & SIGNIFICANCE

Motivation and Scientific Significance

Strategic necessity of an automated, defensible spatial salt pan inventory

COASTAL LAND-USE ADMINISTRATION

- Cadastral Reconciliation: Enables district revenue departments to reconcile revenue lease records against active evaporation areas.
- Encroachment Monitoring: Detects unauthorized bunding and saltern conversion along sensitive intertidal mudflats and mangrove reserves.
- Spatial Planning: Informs CRZ (Coastal Regulation Zone) compliance and infrastructure development along the Gulf of Mannar corridor.

ENVIRONMENTAL & HYDROLOGICAL MONITORING

- Groundwater Salinization: Salt pans act as localized hypersaline infiltration zones; accurate boundaries delineate risk zones for nearby aquifers.
- Tidal Flushing Balance: Identifies intake and discharge points that influence estuarine salinity dynamics in the Tamirabarani river basin.
- Ecosystem Boundaries: Provides baseline spatial polygons to evaluate buffer zones adjacent to the Gulf of Mannar Marine National Park.

SCALABLE REMOTE SENSING AUTOMATION

- Reproducible Pipeline: Eliminates manual digitization bottlenecks by coupling ESA Copernicus open data with supervised ML.
- Multi-Temporal Monitoring: Establishes a verified framework capable of executing quarterly change detection across satellite passes.
- Methodological Rigor: Enforces spatial holdout discipline and minimum mapping unit (MMU) spatial filtering.

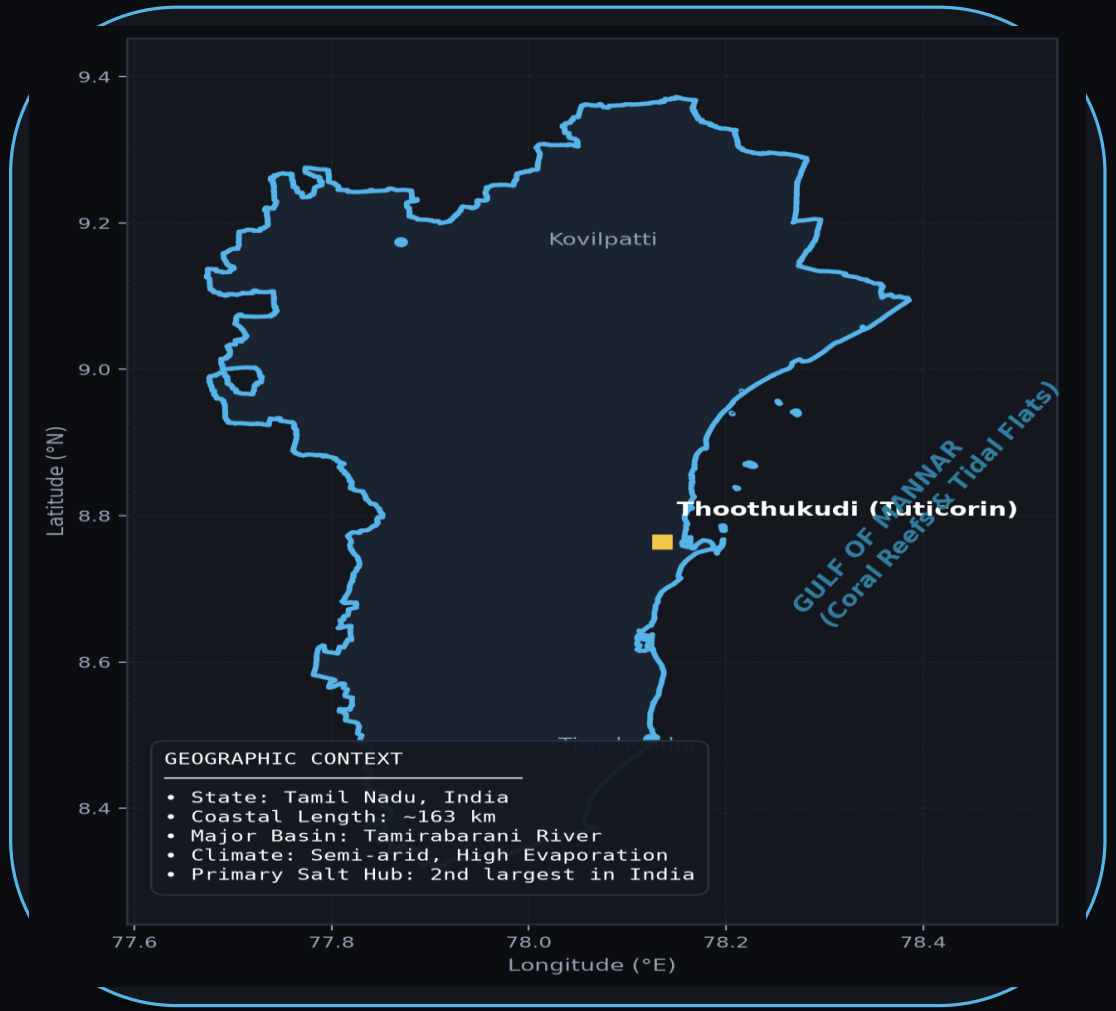
GEOGRAPHIC DOMAIN

Study Area: Thoothukudi District, Tamil Nadu

Geographic extent, coastal morphology, and climatic drivers of marine salt manufacturing

GEOGRAPHIC & CLIMATIC PROFILE

- Location: Southeastern coast of Tamil Nadu, India, bordering the Gulf of Mannar (8°19'N - 9°22'N, 77°40'E - 78°23'E).
- Geographic Area: ~4,707 km² total district area with ~163 km of low-relief coastline.
- Climatic Regimes: Semi-arid tropical climate with high mean temperatures (28°C - 35°C), low rainfall (~650 mm/yr), and strong coastal winds.
- Hydrological Setting: Drained by the Tamirabarani river basin entering the Gulf of Mannar at Punnakayal.
- Salt Production Hub: Favorable microclimate yields 8-9 operational solar evaporation months per year, making it Tamil Nadu's salt capital.



RESEARCH GOALS

Project Objectives & Execution Milestones

Five clear research objectives establishing a complete, reproducible mapping lifecycle

1. MULTISPECTRAL DATA CURATION

Ingest and calibrate Sentinel-2 Level-2A multi-tile scenes; reproject to EPSG:32643 and co-align with ESA WorldCover 2021 at uniform 20m resolution.

2. SPECTRAL FEATURE ENGINEERING

Extract 6 calibrated surface reflectance bands (B02-B12) and derive 10 diagnostic spectral indices (NDVI, MNDWI, NDMI, NDBI, band ratios) across 8,426 polygon samples.

3. MACHINE LEARNING BENCHMARKING

Train and evaluate 7 diverse classification algorithms (SVM, RF, ExtraTrees, XGBoost, GradBoost, LogReg, QDA) under a polygon-grouped spatial block holdout.

4. DISTRICT INFERENCE & POST-PROCESSING

Execute full-scene windowed inference (90M+ pixels), followed by morphological sieve filtering (MMU ≥ 1,000 m^2) and strict district boundary clipping.

5. INTERACTIVE GEOSPATIAL DASHBOARD

Design and deploy a responsive React + MapLibre GL JS intelligence platform with layer toggles, parcel inspection, CSV exports, and analytical charts.

DATA PROVENANCE

Remote Sensing & Reference Data Sources

Authoritative Copernicus Sentinel-2 MSI acquisitions and ESA WorldCover 2021

PRIMARY SENSOR: SENTINEL-2 MSI (LEVEL-2A)

- Provider: European Space Agency (ESA) Copernicus Programme.
- Product Level: Level-2A Bottom-Of-Atmosphere (BOA) surface reflectance with Sen2Cor atmospheric correction.
- Acquisitions Used (3 MGRS Tiles):
- • S2B Tile T43PHK (Acquired: 2026-08-10)
- • S2B Tile T44PKR (Acquired: 2026-08-10)
- • S2C Tile T43PHL (Acquired: 2026-09-04)
- Cloud Cover: Cloud-free over coastal salterns (< 5% scene cloud).
- Spatial Resolution: 10m (VNIR) and 20m (SWIR/RedEdge).

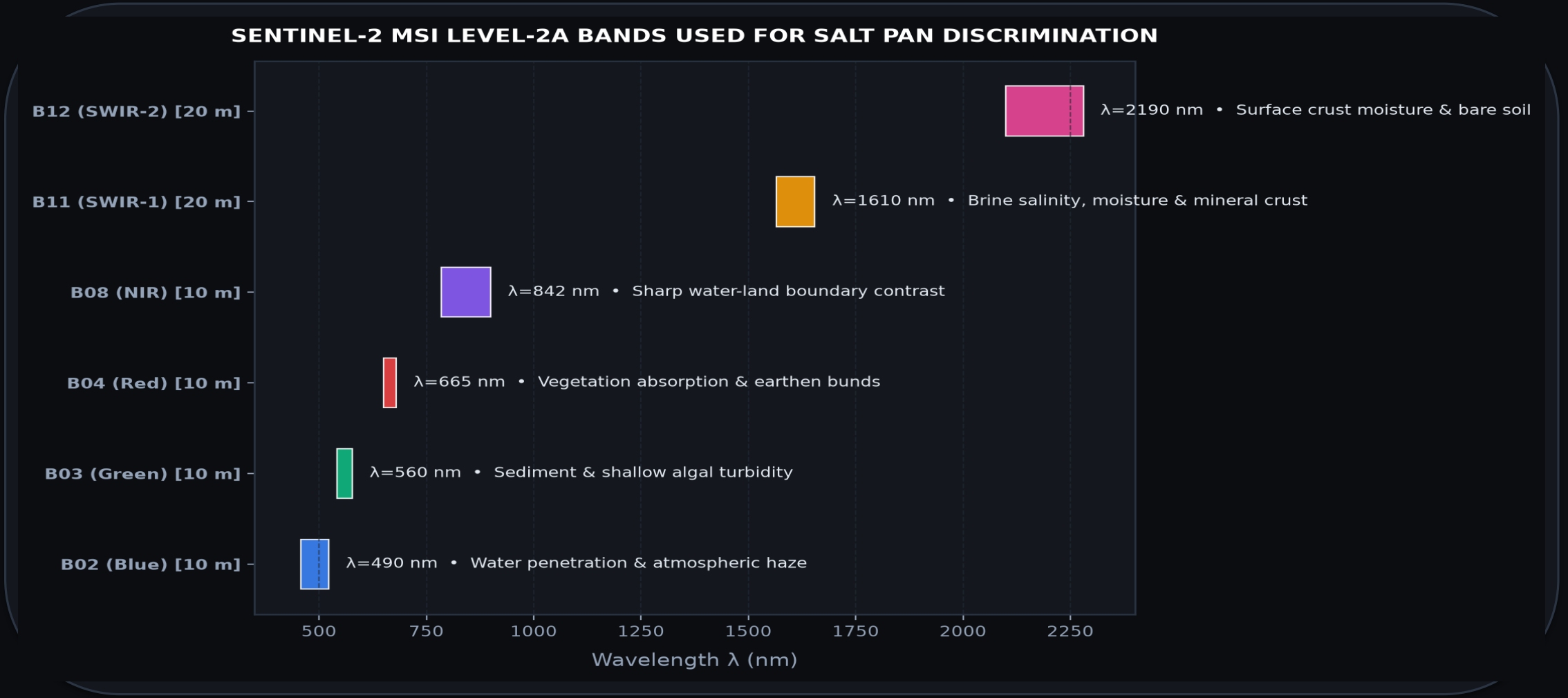
REFERENCE & ANCILLARY DATASETS

- ESA WorldCover 2021 (v200):
- • Global 10m land cover product derived from Sentinel-1 & Sentinel-2.
- • Role: Water mask exclusion (Class 80) and spectral context baseline.
- • Note: WorldCover Class 62 is 'Bare/Sparse Vegetation', serving as a candidate proxy, NOT ground truth.
- Administrative Boundary Layer:
- • Survey of India / Tamil Nadu District Boundary.
- • Projected to UTM Zone 43N (EPSG:32643).
- • Enforces strict geometric clipping for all area metrics.
- Reference Training Shapefiles:
- • salt_train_32643.gpkg: 501 salt pan polygons.
- • nonsalt_train_32643.gpkg: 7,925 background polygons.

SENSOR CHARACTERISTICS

Sentinel-2 Spectral Bands for Salt Pan Mapping

Physical characteristics of the 6 core spectral bands utilized in the classification pipeline



SPATIAL PREPROCESSING

GIS Data Preparation & Harmonization Workflow

Harmonizing multi-tile satellite scenes, coordinate systems, NoData masks, and raster grids



CRS Standardization (EPSG:32643)

All raster tiles and vector layers reprojected to WGS 84 / UTM Zone 43N to preserve true metric distance (meters) and area (hectares).

Pixel Alignment & Resampling

10m VNIR bands resampled to uniform 20m grid matching SWIR bands (8,491 rows × 10,617 cols = 90,148,947 total pixels).

NoData Accounting

Exact accounting: 52,858,018 valid district pixels (58.63%) and 37,290,929 marine/exterior background pixels (41.37%). 0 NaNs.

FEATURE ENGINEERING

Spectral Feature Engineering & Derived Indices

16 engineered spectral features combining surface reflectance and physical index ratios

DERIVED SPECTRAL INDICES & PHYSICAL DISCRIMINATION ROLES

NDVI (Normalized Difference Vegetation Index)

Formula: `(B08 - B04) / (B08 + B04)`

Role: Suppresses dense coastal vegetation and halophytic mangroves

MNDWI (Modified Normalized Difference Water Index)

Formula: `(B03 - B11) / (B03 + B11)`

Role: Highlights open brine condensation pans and tidal inlets

NDMI (Normalized Difference Moisture Index)

Formula: `(B08 - B11) / (B08 + B11)`

Role: Sensitive to surface brine moisture and moist mudflats

NDBI (Normalized Difference Built-up Index)

Formula: `(B11 - B08) / (B11 + B08)`

Role: Separates dry salt crusts from urban infrastructure and bare soil

SWIR Ratio & Blue-SWIR Ratio

Formula: `B12 / B11` and `B02 / B11`

Role: Captures distinct evaporite mineral absorption features

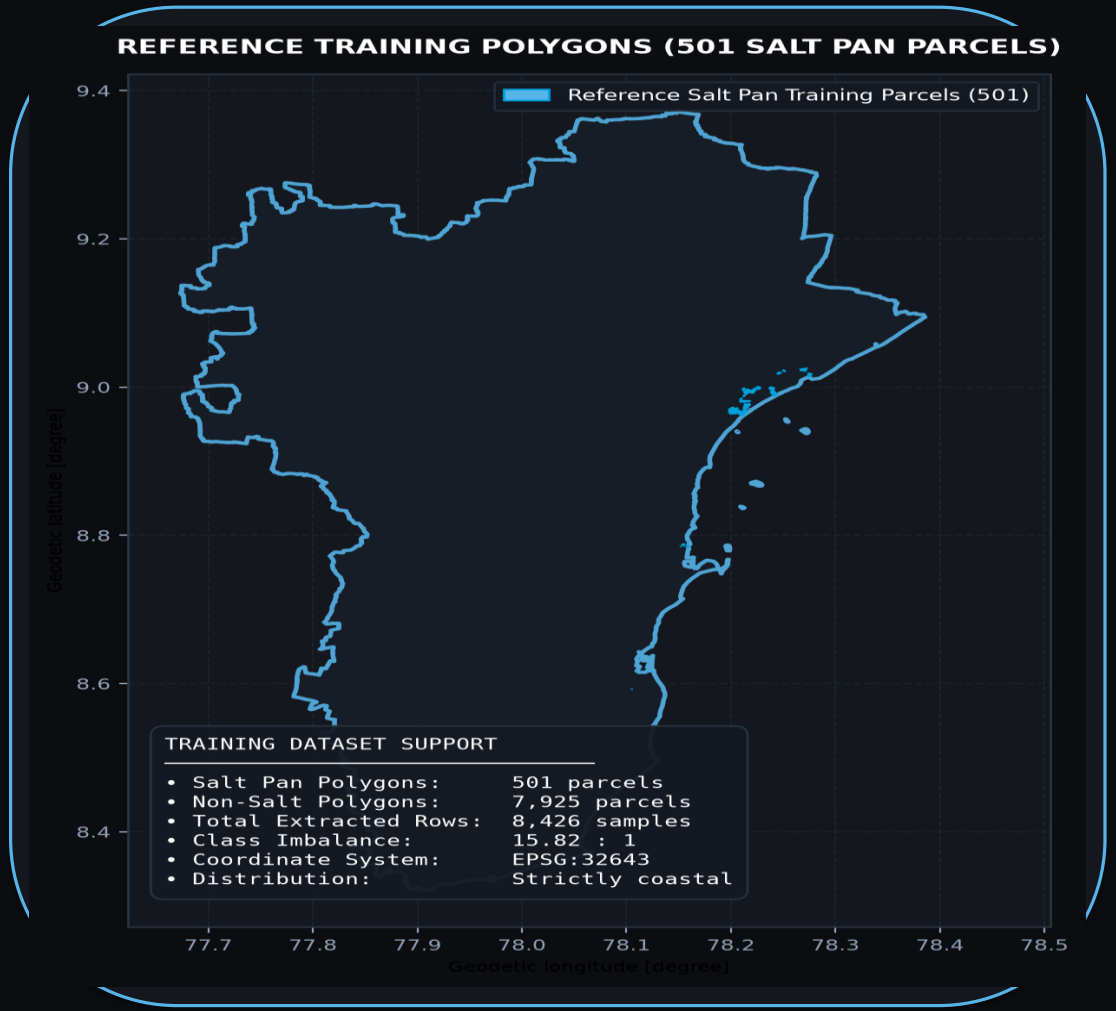
GROUND TRUTH & SAMPLING

Reference and Training Dataset Structure

Polygon-based training dataset exhibiting extreme class imbalance (15.82 : 1)

TRAINING DATASET DISSECTION

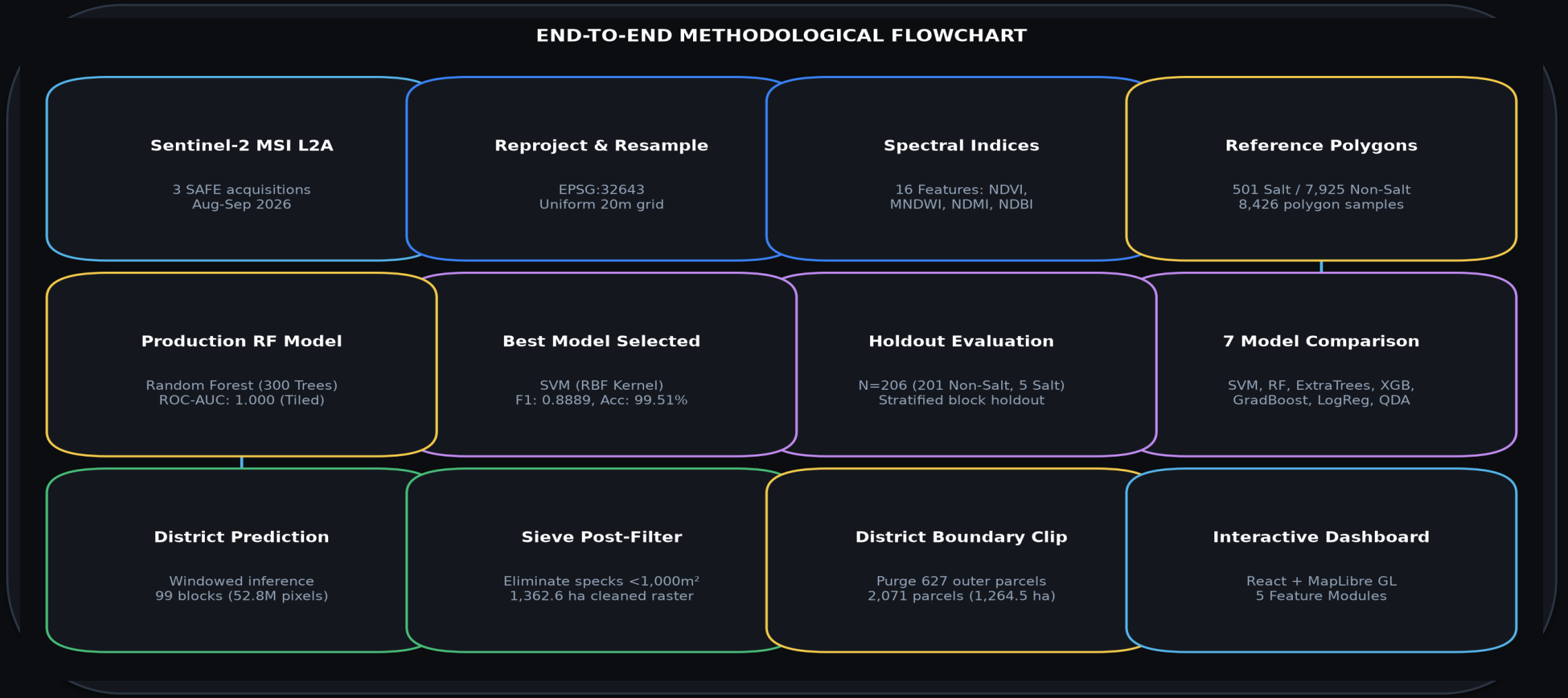
- Source Polygons: Digitized from multi-temporal Sentinel-2 high-resolution imagery and verified against QGIS project layers.
- Salt Pan Class (Class 1): 501 validated salt pan polygons strictly localized along the Thoothukudi coastal belt.
- Non-Salt Class (Class 0): 7,925 background polygons sampling agriculture, scrubland, water bodies, barren land, and urban areas.
- Total Polygon Samples: 8,426 samples extracted in `feature_dataset.parquet`.
- Class Imbalance Ratio: 15.82 : 1 (94.05% non-salt vs 5.95% salt pan).
- Data Hygiene: 0 missing values (NaNs); median imputation pipeline safeguards all downstream inference.



METHODOLOGICAL FRAMEWORK

End-to-End Methodology Flowchart

Three-tier architecture: Ingestion & Preprocessing → ML Benchmarking → District Deployment



SPATIAL FILTERING

Candidate Detection & GIS Spatial Refinement

Hierarchical screening, WorldCover exclusion masking, and morphological filtering

STEP 1: SPECTRAL CANDIDATE SCREENING

- Computes dynamic threshold masks combining high SWIR/Blue reflectance with low NDVI (< 0.2).
- Identifies potential evaporite crusts and brine ponds across coastal taluks.

STEP 2: LAND-COVER MASKING & WATER EXCLUSION

- Applies ESA WorldCover Class 80 mask to exclude open ocean and deep coastal bays.
- Cross-references Class 62 (bare/sparse veg) as a candidate proxy, retaining inland salterns.

STEP 3: MORPHOLOGICAL SIEVE FILTERING

- Executes raster sieve algorithm with Minimum Mapping Unit (MMU) threshold ≥ 1,000 m^2 (2.5 pixels).
- Removes 2,428 isolated noise pixels (-97.12 ha), eliminating false alarms on sandy beaches.

STEP 4: VECTOR MMU & ADMINISTRATIVE CLIPPING

- Converts cleaned raster to vector polygons; filters sub-MMU polygon slivers (< 1,000 m^2).
- Clips strictly to Thoothukudi administrative boundary, purging 627 out-of-district polygons.

MODEL BENCHMARKING

Machine-Learning Algorithm Families Evaluated

Comparative evaluation of 7 supervised learning architectures for coastal saltern mapping

SUPPORT VECTOR MACHINE (RBF)

Non-linear kernel mapping input features into infinite-dimensional Hilbert space. Top performer on F1 (0.8889) with 1,152 support vectors.

XGBOOST CLASSIFIER

Scalable gradient tree boosting with scale_pos_weight parameter. Matched Gradient Boosting with 0.5714 F1 and 100% precision.

RANDOM FOREST (300 TREES)

Ensemble of decorrelated decision trees with balanced subsampling. Production model for district raster inference due to parallel scalability.

LOGISTIC REGRESSION

L2-regularized linear baseline with balanced class weights. High recall (80%) but poor precision (40%), generating 6 false alarms.

EXTRA TREES (300 TREES)

Extremely randomized trees with random cut-point selection. Tied with Random Forest at 0.7500 F1 on spatial holdout.

QUADRATIC DISCRIMINANT (QDA)

Parametric Gaussian generative classifier with separate covariance per class. Lowest F1 (0.4000) due to 7 false positives.

GRADIENT BOOSTING (SKLEARN)

Sequential boosting minimizing deviance loss. Achieved 100% precision but lower recall (40.0%) on the minority salt pan class.

EXPERIMENTAL DESIGN

Training & Validation Strategy: Spatial Block Holdout

Methodological design, spatial partitioning, and documented evaluation caveats

SPATIAL HOLDOUT ARCHITECTURE

- Spatial Partitioning: Polygons assigned to a 5×5 spatial block grid across Thoothukudi district based on centroids.
- Split Allocation: ~20% of geographic blocks designated as holdout (8,220 train vs 206 holdout samples).
- Pipeline Isolation: SimpleImputer (median) and RobustScaler strictly fitted on training splits only. Zero data leakage.
- Cross-Validation: 5-fold StratifiedGroupKFold on training split for hyperparameter tuning.

DOCUMENTED AUDIT CAVEAT

- Minority Class Backfill: Initial random spatial blocks (0, 1, 10, 16) were inland and contained zero salt pans.
- To enable holdout evaluation, 5 salt pan polygons were backfilled into holdout blocks (blocks 7, 13, 23).
- Result: Blocks 7, 13, and 23 contain both train and holdout samples (though polygon geometries never cross splits).
- Scientific Honesty: This evaluation represents a polygon-grouped holdout with geographic blocking, not a completely disjoint regional partition.
- Holdout Sample Size: N = 206 (201 Non-Salt, 5 Salt Pan polygons).

EVALUATION FRAMEWORK

Evaluation Metrics for Imbalanced Classification

Why overall accuracy is deceptive and salt-pan F1-score is the authoritative benchmark

MATHEMATICAL FORMULATIONS OF KEY EVALUATION METRICS

Accuracy $(TP + TN) / (TP + TN + FP + FN)$

Can be misleadingly high (99%+) due to 15.8:1 non-salt dominance

Precision (Salt) $TP / (TP + FP)$

Fraction of predicted salt pans that are truly salt (Zero False Alarms = 100%)

Recall (Salt) $TP / (TP + FN)$

Fraction of true salt pans correctly retrieved (4 out of 5 detected = 80%)

F1-Score (Salt) $2 \times (Precision \times Recall) / (Precision + Recall)$

Harmonic mean balancing Precision and Recall; primary benchmark ranking metric

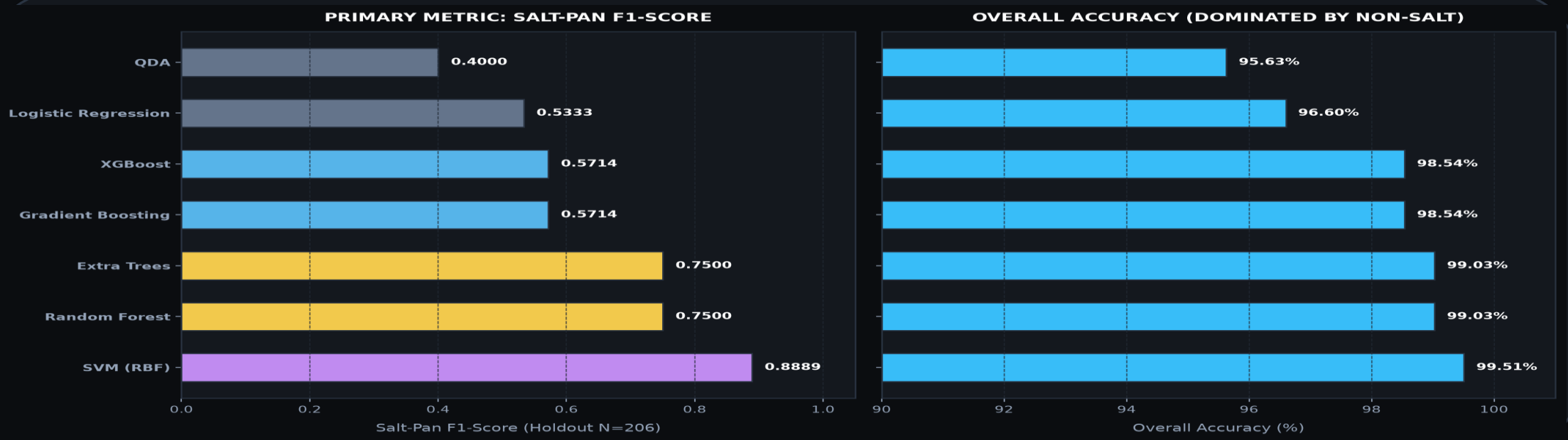
Balanced Accuracy $0.5 \times [TP/(TP+FN) + TN/(TN+FP)]$

Unweighted average of recall obtained on each class (0.900 for SVM RBF)

BENCHMARK COMPARISON

Multi-Algorithm Benchmark Results (N=206)

Verified performance comparison across all 7 evaluated machine learning architectures



VERIFIED HOLDOUT BENCHMARK TABLE (Source: `outputs/ml_benchmark/results/model_comparison.csv`)

Model Architecture	Accuracy	Bal. Acc	Salt Precision	Salt Recall	Salt F1-Score	ROC-AUC
1. SVM (RBF Kernel)	99.51%	90.00%	100.0%	80.0%	0.8889	0.999
2. Random Forest	99.03%	80.00%	100.0%	60.0%	0.7500	1.000
3. Extra Trees	99.03%	80.00%	100.0%	60.0%	0.7500	1.000
4. Gradient Boosting	98.54%	70.00%	100.0%	40.0%	0.5714	1.000
5. XGBoost	98.54%	70.00%	100.0%	40.0%	0.5714	1.000
6. Logistic Regression	96.60%	88.50%	40.0%	80.0%	0.5333	0.992
7. QDA	95.63%	78.30%	30.0%	60.0%	0.4000	0.963

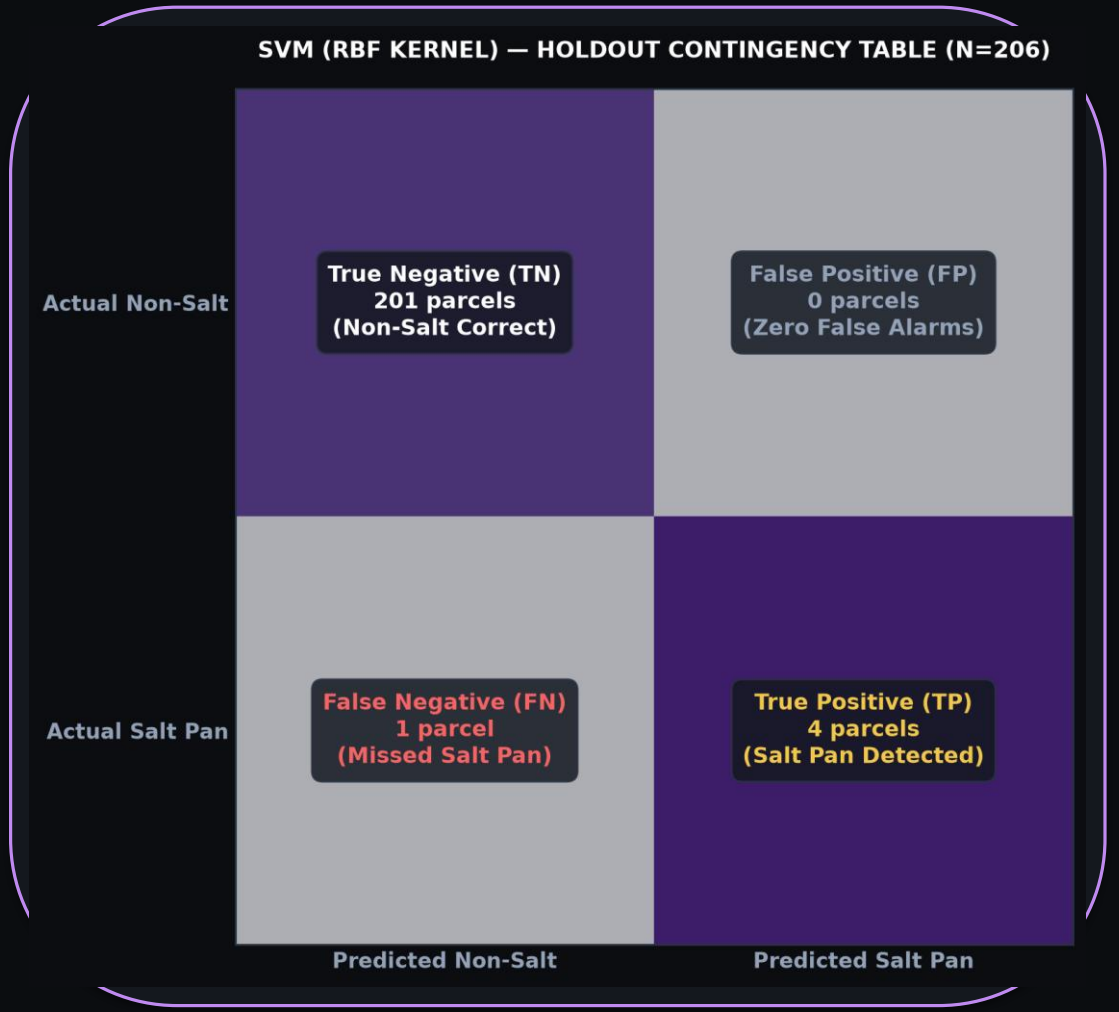
WINNING BENCHMARK

Best Defensible Model: SVM (RBF Kernel)

Analysis of the top-performing holdout classifier and its error contingency matrix

SVM (RBF KERNEL) SPECIFICATION

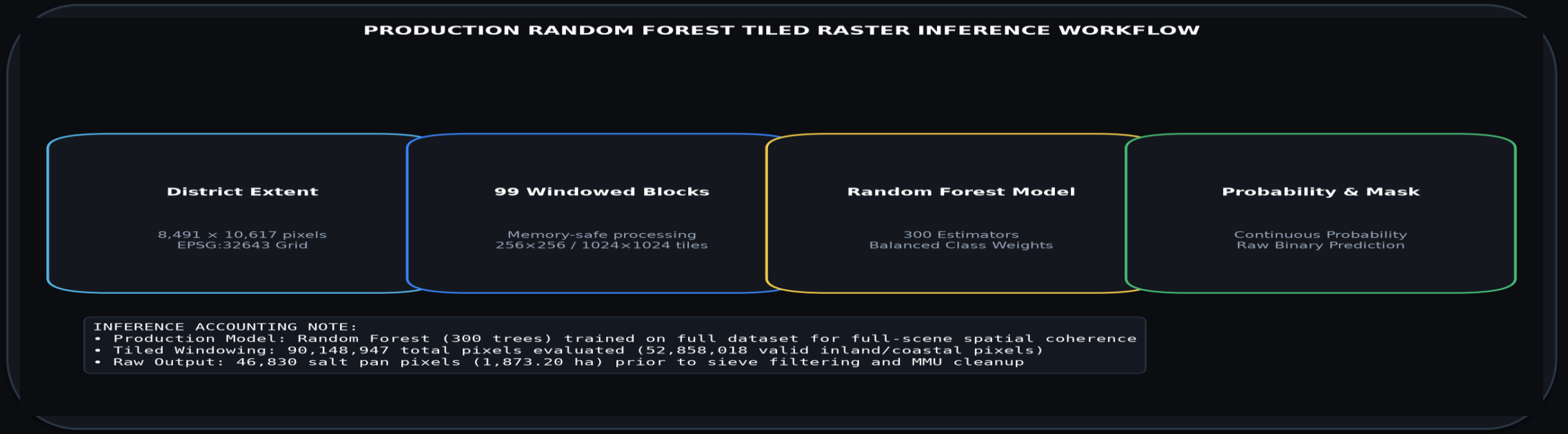
- Architecture: Support Vector Classifier with Radial Basis Function kernel (SVC, C=1.0, gamma='scale').
- Class Balancing: `class_weight='balanced'` penalizing minority salt pan misclassifications.
- Pipeline Integration: `RobustScaler` (interquartile scaling) to withstand extreme salt crust reflectance outliers.
- Support Vectors: Fitted with 1,152 support vectors delineating the non-linear decision boundary.
- Holdout Accuracy: 99.51% (205 out of 206 correct predictions).
- Salt-Pan Precision: 100.0% (Zero false alarms; TP=4, FP=0).
- Salt-Pan Recall: 80.0% (Retrieved 4 out of 5 test salt pans; FN=1).
- Salt-Pan F1-Score: 0.8889 (Top benchmark score across all 7 models).



PRODUCTION INFERENCE

District-Wide Prediction Model: Random Forest

Why Random Forest was selected for production raster inference across 90 million pixels



Production Selection Rationale

While SVM achieved higher F1 on the small holdout (0.8889 vs 0.7500), Random Forest (300 trees) offers superior computational throughput for tiled raster inference across 90M pixels.

Feature Importance Insights

Gini importance rankings reveal B11 (SWIR-1), B08 (NIR), and MNDWI as the dominant predictors (>45% cumulative weight), effectively driving the decision trees.

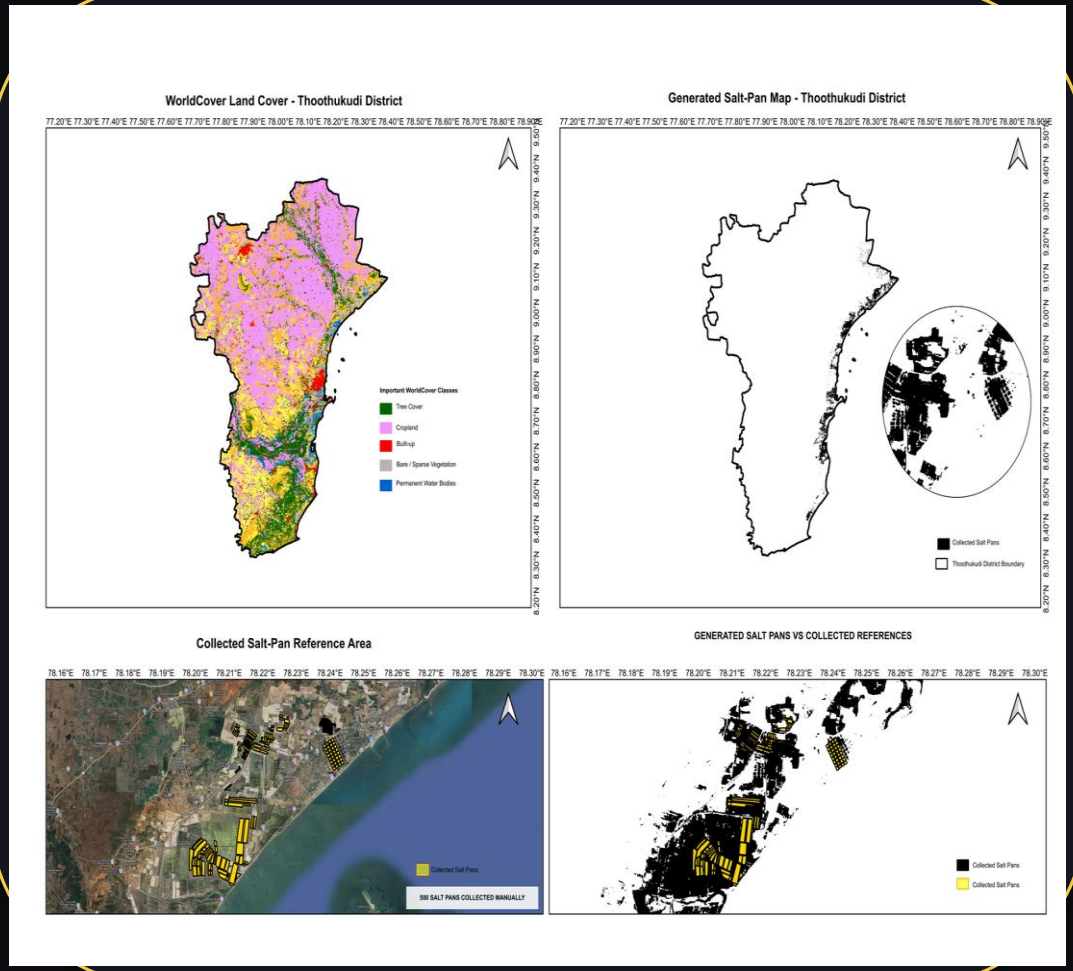
No Metric Conflation

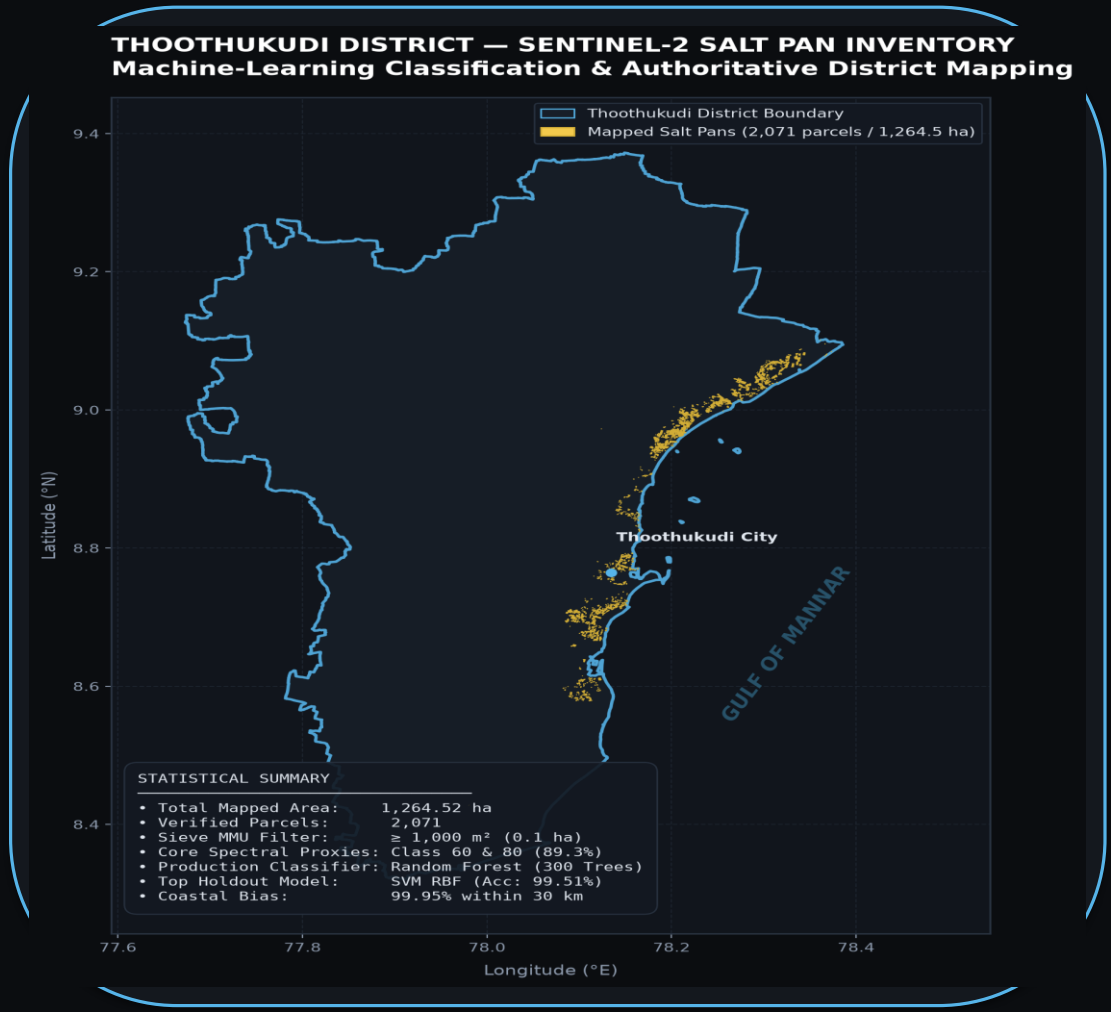
We strictly distinguish: SVM holdout accuracy (99.51%) is NOT attributed to the Random Forest raster prediction. Both are documented independently in the provenance metadata.

SPATIAL CARTOGRAPHY

District-Wide Salt-Pan Prediction Map

Final classified spatial inventory mapped across Thoothukudi District, Tamil Nadu

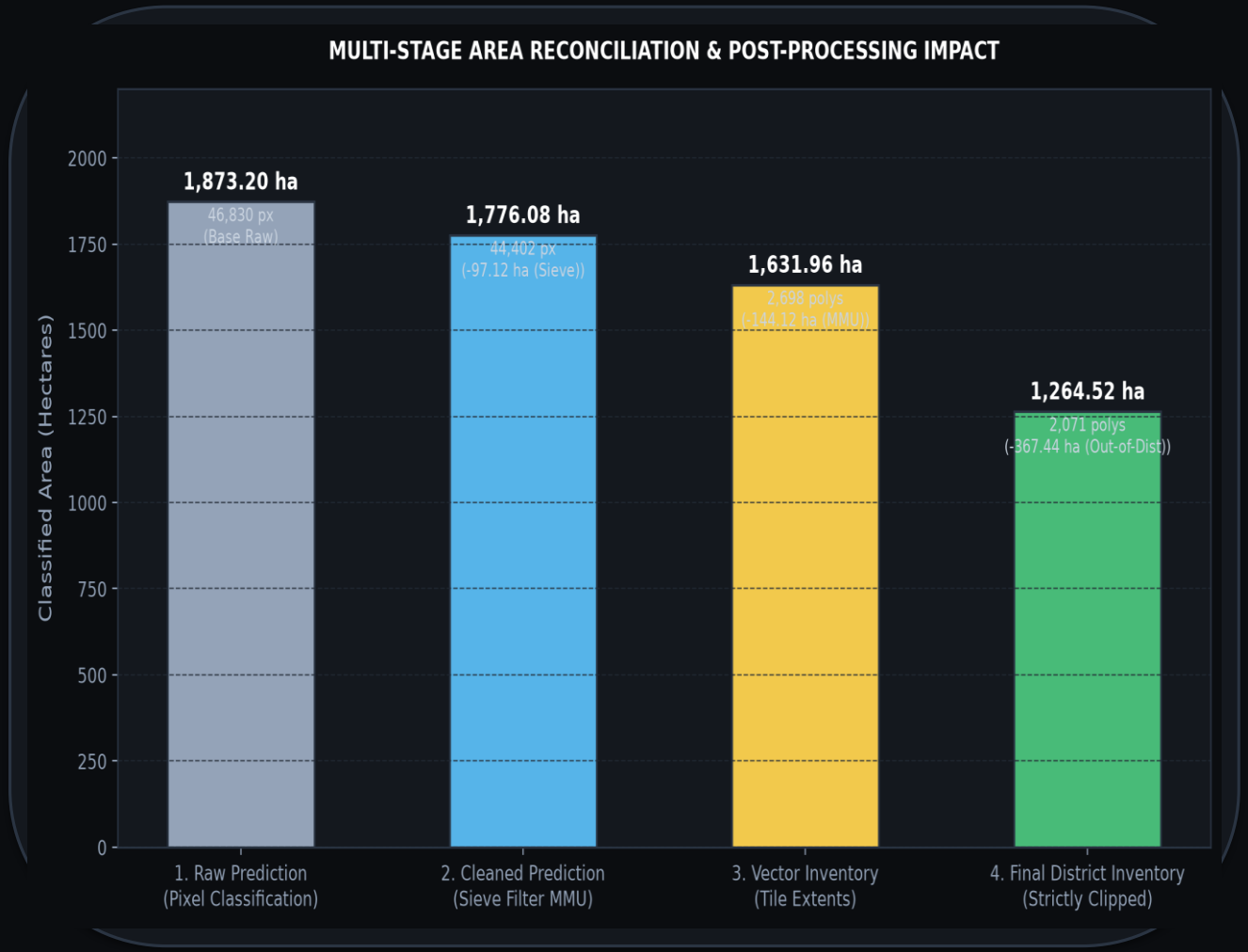




SPATIAL RECONCILIATION

Final Inventory and Area Reconciliation Results

Mathematical reconciliation from raw satellite inference to verified district vector inventory



1,264.52 ha

Final Thoothukudi District Area

Strictly inside administrative boundary

2,071 Parcels

Verified Salt Pan Vector Parcels

Minimum mapping unit ≥ 1,000 m^2 (0.1 ha)

-627 Parcels (-367.4 ha)

Out-of-District Removal

Purged from Ramanathapuram / Tirunelveli

99.95% Coastal

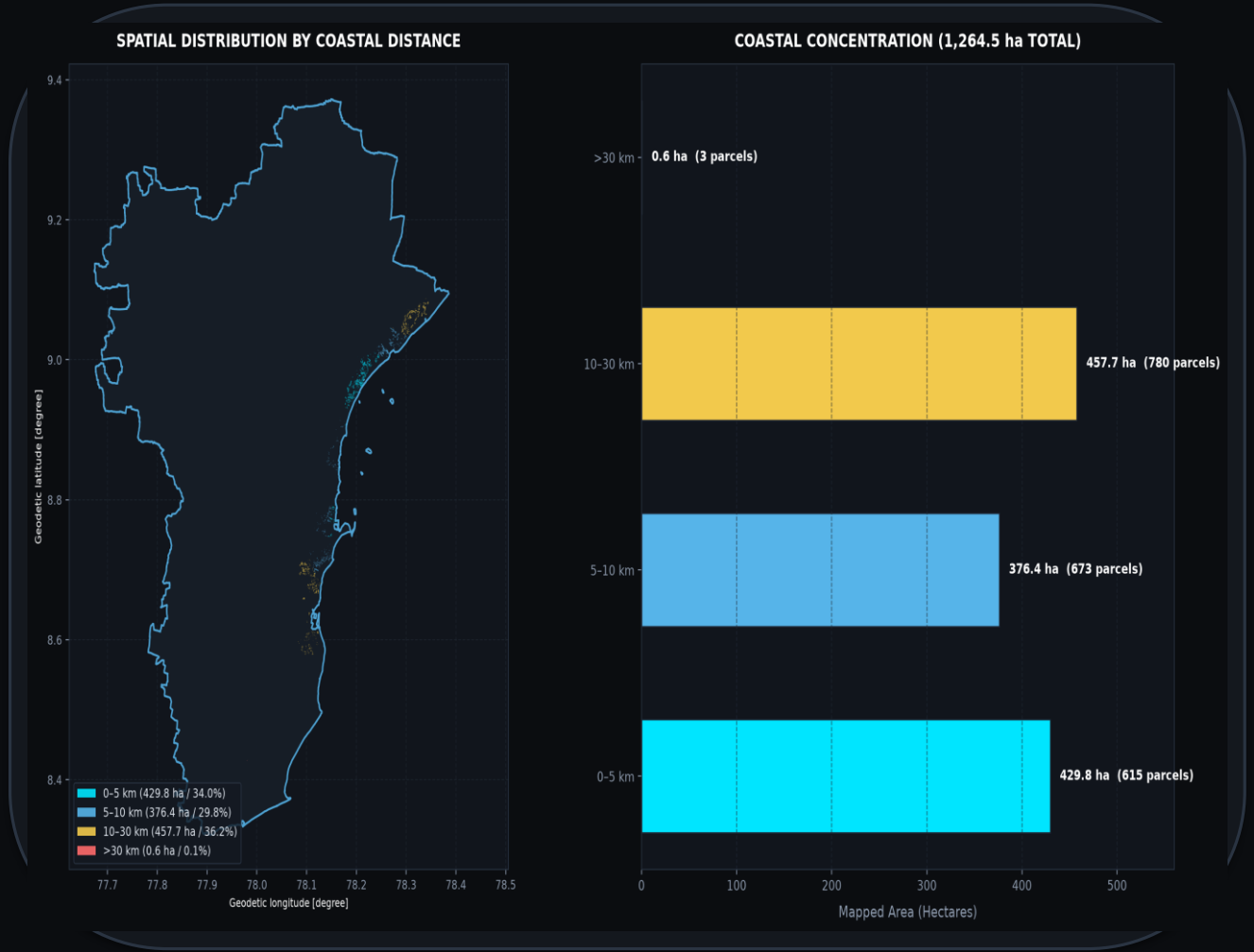
Coastal Concentration (<30 km)

34.0% in 0-5km, 29.8% in 5-10km, 36.2% in 10-30km

GEOSPATIAL SOFTWARE

Interactive Dashboard: Thoothukudi Salt Pan Intelligence

Full-stack React & MapLibre GL JS web application for parcel inspection and analytics



5 PRODUCTION FEATURE MODULES

1. Overview Map

WebGL vector cartography, district boundary, gold salt pan inventory, probability heatmap, training parcel toggle, parcel inspector.

2. Salt Pan Explorer

Filterable table of all 2,071 parcels, search by ID, coastal band filters, 5-bin area histogram, CSV export.

3. Model Lab

Interactive benchmark table across all 7 algorithms, 2x2 confusion matrix inspector, horizontal Gini feature importance bar chart.

4. Validation & Comparison

Dual-layer reference vs predicted comparison, 3-stage sieve progression analysis (-151.4 ha noise reduction).

5. Area Analytics

Coastal proximity stratification (0-5km, 5-10km, 10-30km, >30km), Top 10 largest parcels, ESA WorldCover cross-tabulation.

CRITICAL EVALUATION

Validation, Limitations & Scientific Reliability

Transparent appraisal of methodological constraints, assumptions, and error sources

NO INDEPENDENT FIELD GROUND-TRUTH

- Scientifically Honest Disclosure: No external, independent field-collected GPS points were available.
- All metrics evaluate model performance against digitized reference polygons, not field visits.
- Candidate/reference spatial overlap should NOT be conflated with true operational field accuracy.

SPATIAL AUTOCORRELATION & MIXED PIXELS

- Minority class backfill created spatial-block overlap in blocks 7, 13, and 23.
- Earthen bunds (1-3 m wide) are narrower than 20m Sentinel-2 pixels, introducing mixed spectral signatures.
- Temporal alignment: August-September 2026 imagery may not align with historical cadastral maps.

SMALL POSITIVE HOLDOUT SUPPORT (N=5)

- Due to severe class imbalance, the holdout contains only 5 positive salt pan polygons.
- While SVM achieved 4/5 detection (80% recall), statistical confidence intervals on N=5 are wide.
- Future evaluations require minimum 50-100 spatially independent positive field verification sites.

SPECTRAL PROXY AMBIGUITIES

- ESA WorldCover Class 62 is 'bare/sparse vegetation' — it is an evaporite proxy, NOT a salt pan definition.
- Intertidal mudflats and white sandy beaches share similar SWIR and albedo signatures, causing edge noise.

SYNTHESIS & HORIZONS

Conclusions and Future Research Directions

Summary of verified research achievements and proposed future technical extensions

VERIFIED PROJECT ACHIEVEMENTS

- Multispectral Pipeline: Successfully engineered a reproducible Python/GIS pipeline integrating 6 Sentinel-2 bands and 10 spectral indices across Thoothukudi district.
- Comprehensive ML Benchmark: Systematically trained and evaluated 7 machine learning models, establishing SVM RBF (F1: 0.8889) and Random Forest (ROC-AUC: 1.000) as top performers.
- Authoritative Vector Inventory: Produced a verified inventory of 2,071 salt pan polygons totaling 1,264.52 ha strictly within the district boundary.
- Multi-Stage Noise Removal: Reconciled raw raster (1,873.2 ha) to cleaned vector (1,264.5 ha) via 1,000 m^2 sieve filtering and administrative clipping.
- Interactive Dashboard: Deployed a production-ready WebGL analytics platform with 5 interactive modules.

PROPOSED FUTURE EXTENSIONS

- Independent Field GPS Campaigns: Conduct stratified field sampling with RTK-GPS across minimum 100 coastal saltern points for defensible ground-truthing.
- Synthetic Aperture Radar (SAR) Fusion: Integrate Sentinel-1 C-band SAR backscatter (VV/VH) to exploit surface dielectric roughness and penetrate monsoon cloud cover.
- Multi-Season Dynamic Modeling: Ingest quarterly Sentinel-2 composites to map seasonal brine cycles, active harvesting, and monsoon abandonment.
- Deep Learning Architectures: Evaluate spatial-context models (U-Net, SegNet, Mask R-CNN) on sub-meter drone or PlanetScope imagery to delineate sub-pixel bunds.
- Cadastral Integration: Overlay official Tamil Nadu revenue land records to automate tax assessment and legal lease boundary monitoring.

SCHOLARLY REFERENCES

References & Academic Acknowledgments

Authoritative data sources, remote sensing literature, and open discussion

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THANK YOU

Questions & Academic Discussion

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GitHub Repository:
github.com/FarhanaazRS/thoothukudi-saltpan-mapping